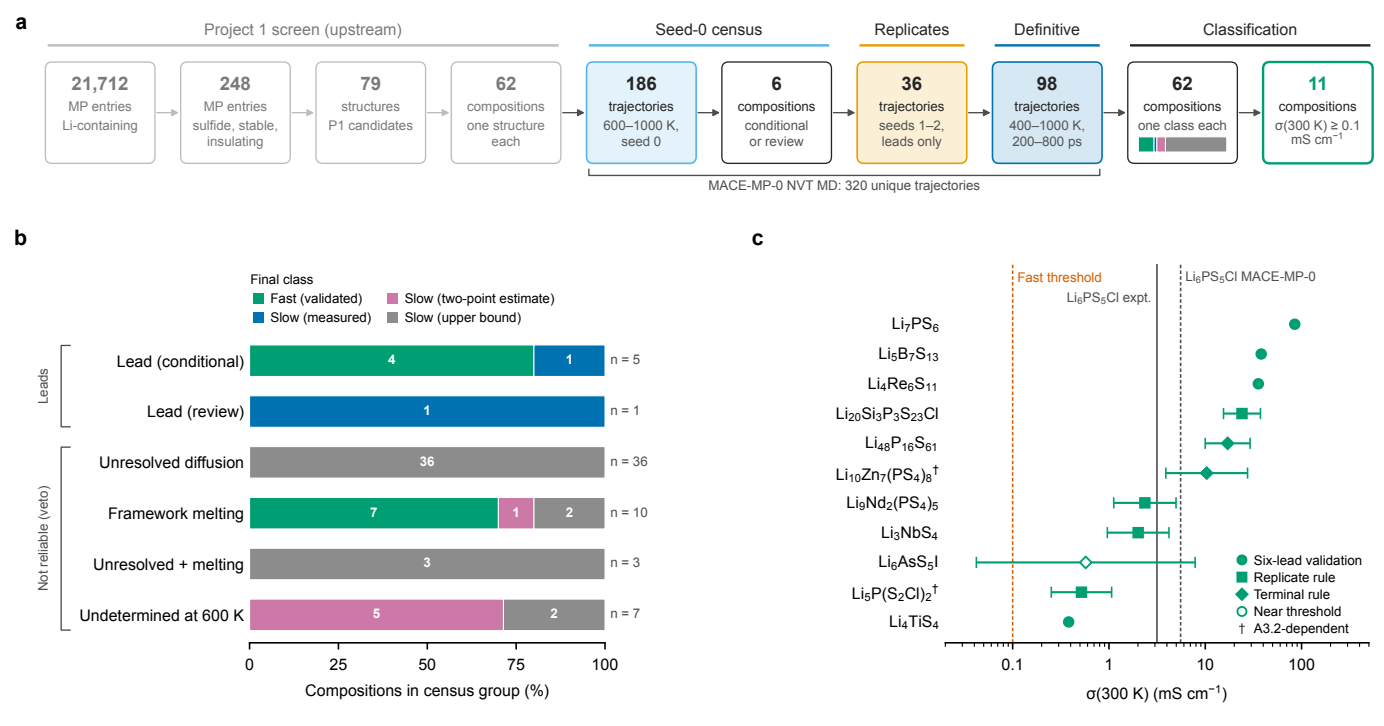

a
Project 1 screen (upstream)
Seed-0 census
Replicates
Definitive
Classification
21,712
MP entries
Li-containing
248
MP entries
sulfide, stable, insulating
79
structures
P1 candidates
62
compositions
one structure each
186
trajectories
600–1000 K, seed 0
6
compositions
conditional or review
36
trajectories
seeds 1–2, leads only
98
trajectories
400–1000 K, 200–800 ps
62
compositions
one class each
11
compositions
σ(300 K) ≥ 0.1 mS cm−1
MACE-MP-0 NVT MD: 320 unique trajectories
b
Final class
Fast (validated)
Slow (two-point estimate)
Slow (measured)
Slow (upper bound)
Leads
Lead (conditional)
Lead (review)
Not reliable (veto)
Unresolved diffusion
Framework melting
Unresolved + melting
Undetermined at 600 K
n = 5
n = 1
n = 36
n = 10
n = 3
n = 7
0
25
50
75
100
Compositions in census group (%)
c
Fast threshold
Li6PS5Cl expt.
Li6PS5Cl MACE-MP-0
Li7PS6
Li5B7S13
Li4Re6S11
Li20Si3P3S23Cl
Li48P16S61
Li10Zn7(PS4)8†
Li9Nd2(PS4)5
Li3NbS4
Li6AsS5I
Li5P(S2Cl)2†
Li4TiS4
Six-lead validation
Replicate rule
Terminal rule
Near threshold
† A3.2-dependent
0.1
1
10
100
σ(300 K) (mS cm−1)